

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0179

Centerville Sw 100 kV
January 23, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0179. The Interconnection Customer proposes a 233.8 MW wind facility interconnecting at the Centerville Sw 100 kV substation in the Ohio area. The request was received on November 26, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 137 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$158,347,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wilsall to Hebron 100 KV	Denmark to Centerville Sw 100 KV	203.2	106	100.0
Wilsall to Centerville Sw 100 KV	Centerville Sw to Denmark 100 KV	193.8	117	100.0
Centerville Sw to Denmark 100 KV	Centerville Sw to Wilsall Substation 100 KV	187.8	120	100.0
Pockrus to Hebron 100 KV L	Centerville Sw Substation to Denmark 100 KV	179.5	117	100.0
Denmark to Thrive Fontana 100 KV	Watermill to Liberty 100 KV L	155.5	130	79.8
Pockrus to Prosser 100 KV line	Denmark to Thrive Fontana 100 KV	148.3	122	100.0
Adams to Onancock Diesel Generation Facility 230 KV	Trego to Beaver 500 KV Line	144.2	670	6.7
Liberty to Watermill 100 KV L	Denmark to Thrive Fontana 100 KV	142.0	120	61.5
Post Rock to Newport Tie 230 KV	Trego to Beaver 500 KV	138.3	543	21.4
Santa Isabel to Pockrus 100 KV	Centerville Sw Substation to Denmark 100 KV Line	130.8	105	51.7
Post Rock to Myrle Point 230 KV Line	Trego to Beaver 500 KV	130.4	621	16.7
Prosser to Watermill 100 KV Line	Thrive Fontana to Denmark 100 KV line	130.0	129	100.0
Joso to Onancock Diesel Generation Facility 230 KV	Eufaula to Beaver 500 KV line	128.7	537	6.4
Colleen to Liberty 100 KV	Denmark to Thrive Fontana 100 KV Line	119.9	128	61.5
Beaver 230 kV/1 kV Autotransformer	Beaver to Eufaula 500 KV	115.5	1,021	6.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Canal Rd Substation to Wedpatch 230 KV	Beaver to Trego 500 KV L	113.2	394	13.8
Santa Isabel to Denmark 100 KV line	Centerville Sw to Denmark 100 KV	112.7	116	51.7
Beaver Substation to Onancock Diesel Generation Facility 230 KV	Beaver to Eufaula 500 KV	112.4	705	5.1
Mcgill to Clender 100 KV	Post Rock to Myrtle Pt 230 KV	110.7	122	8.6
Beaver 500 kV/1 kV Auto	Beaver to Eufaula 500 KV	107.6	1,096	6.2
Youngwood to Woodland Tap 345 KV	Allegheny Enegy Units 3 4 & 5 to Southeast Vicksburg 500 KV	106.4	1,221	6.4
Ritter Park to Sugar Hill 345 KV	Hilltop to Piney Creek Project 500 KV L	104.7	1,038	11.4
Allegheny Energy Units 3 4 & 5 345/1 kV XFMR	Hilltop to Piney Creek Project 500 KV line	104.0	796	10.8
Gavin Power Llc to Olympus 230 KV	Gavin Power Llc 500 kV/345 kV XFMR	103.7	512	5.8
Montour Sta 500-345 kV Transformer	Westwood Generation Llc to Coronet 500 KV line	101.2	759	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wilsall to Hebron 100 KV	Rebuild line with higher-capacity conductor	\$6,313,000	100% (\$6,313,000)
Wilsall to Centerville Sw 100 KV	Rebuild line with higher-capacity conductor	\$3,714,000	100% (\$3,714,000)
Centerville Sw to Denmark 100 KV	Rebuild line with higher-capacity conductor	\$2,548,000	100% (\$2,548,000)
Pockrus to Hebron 100 KV L	Rebuild line with higher-capacity conductor	\$4,988,000	100% (\$4,988,000)
Denmark to Thrive Fontna Substation 100 KV	Rebuild line with higher-capacity conductor	\$4,931,000	100% (\$4,931,000)
Pockrus to Prosser 100 KV line	Rebuild line with higher-capacity conductor	\$5,230,000	100% (\$5,230,000)
Onancock Diesel Generation Facility to Adams 230 KV Line	Rebuild line with higher-capacity conductor	\$64,280,000	19% (\$12,400,000)
Liberty to Watermill 100 KV L	Rebuild line with higher-capacity conductor	\$5,249,000	100% (\$5,249,000)
Post Rock to Newport Tie 230 KV	Rebuild line with higher-capacity conductor	\$38,338,000	44% (\$17,069,000)
Santa Isabel to Pockrus 100 KV	Rebuild line with higher-capacity conductor	\$5,515,000	100% (\$5,515,000)
Post Rock to Myrle Point 230 KV Line	Rebuild line with higher-capacity conductor	\$20,664,000	40% (\$8,222,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Prosser to Watermill 100 KV Line	Rebuild line with higher-capacity conductor	\$2,055,000	100% (\$2,055,000)
Joso to Onancock Diesel Generation Facility 230 KV	Rebuild line with higher-capacity conductor	\$25,889,000	20% (\$5,151,000)
Liberty Substation to Colleen 100 KV	Rebuild line with higher-capacity conductor	\$4,401,000	100% (\$4,401,000)
Beaver 230-1 kV Autotransformer	Replace transformer with higher-rated unit	\$13,750,000	20% (\$2,728,000)
Canal Rd Substation to Wedpatch 230 KV	Rebuild line with higher-capacity conductor	\$11,801,000	46% (\$5,393,000)
Denmark to Santa Isabel 100 KV	Rebuild line with higher-capacity conductor	\$3,146,000	100% (\$3,146,000)
Beaver to Onancock Diesel Generation Facility 230 KV	Rebuild line with higher-capacity conductor	\$16,906,000	20% (\$3,421,000)
Mcgill to Clender 100 KV	Rebuild line with higher-capacity conductor	\$6,746,000	100% (\$6,746,000)
Beaver 500 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$48,391,000	20% (\$9,601,000)
Youngwood to Woodland Tap 345 KV	Reconductor line	\$25,249,000	8% (\$2,075,000)
Ritter Park to Sugar Hill 345 KV	Reconductor line	\$7,790,000	9% (\$687,000)
Allegheny Energy Units 3 4 & 5 345/1 kV XFMR	Replace transformer with higher-rated unit	\$17,567,000	9% (\$1,544,000)
Olympus to Gavin Power Llc 230 KV L	Reconductor line	\$3,250,000	6% (\$207,000)
Montour Station Substation 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$35,013,000	100% (\$35,013,000)
	Total allocated to project		\$158,347,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.